

## TUXTLA GUTIERREZ, Mexico

WMO: 768430

Lat: 16.7631N

Lon: 93.1475W

Elev: 1890

StdP: 13.72

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 55.2       | 57.4 | 48.0                          | 53.2 | 71.0 | 50.3 | 57.9 | 70.0 | 33.1                  | 69.5 | 29.7 | 67.6 | 10.0                     | 0    | 0.314 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 23.7                   | 98.4            | 72.2 | 95.4 | 72.2 | 93.3 | 72.5 | 78.0                | 88.8 | 76.9 | 88.1 | 76.0 | 86.8 | 8.1                  | 320  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 75.1                            | 141.4 | 84.2 | 73.8 | 135.0 | 81.8 | 73.2 | 132.2 | 80.9 | 43.0          | 88.7 | 41.8 | 88.3 | 40.9 | 86.6 | 84.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 29.0              | 23.5 | 21.2 | 47.7                       | 102.6 | 2.9                | 2.2 | 45.6                                               | 104.2 | 43.9       | 105.5 | 42.2       | 106.8 | 40.1       | 108.4 |
|                   |      |      | 46.1                       | 81.8  | 3.2                | 2.5 | 43.8                                               | 83.6  | 42.0       | 85.0  | 40.2       | 86.4  | 37.9       | 88.2  |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 76.6   | 70.8  | 73.8  | 77.2  | 80.6  | 81.5  | 78.9  | 78.3  | 78.5  | 77.7  | 76.5  | 73.8  | 71.9  |
|                                                                        | DBStd       | 5.00   | 4.44  | 4.96  | 4.96  | 4.36  | 3.96  | 3.12  | 2.74  | 2.48  | 2.79  | 3.44  | 3.54  | 3.96  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 22     | 9     | 5     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 5     |
|                                                                        | CDD50       | 9718   | 643   | 666   | 842   | 919   | 976   | 866   | 877   | 885   | 830   | 821   | 713   | 680   |
|                                                                        | CDD65       | 4264   | 187   | 250   | 379   | 469   | 511   | 416   | 412   | 420   | 380   | 356   | 264   | 220   |
|                                                                        | CDH74       | 32571  | 1205  | 2001  | 3361  | 4613  | 4898  | 3065  | 3042  | 2978  | 2334  | 2180  | 1512  | 1382  |
|                                                                        | CDH80       | 12973  | 426   | 878   | 1578  | 2259  | 2269  | 1096  | 1084  | 1034  | 745   | 687   | 452   | 465   |
| (14) Wind                                                              | WSAvg       | 7.2    | 8.9   | 9.1   | 9.4   | 8.7   | 7.4   | 5.2   | 5.4   | 5.5   | 5.2   | 6.4   | 7.7   | 7.5   |
| (15) Precipitation                                                     | PrecAvg     | 36.8   | 0.1   | 0.1   | 0.1   | 0.7   | 2.6   | 8.9   | 7.1   | 6.8   | 7.9   | 2.2   | 0.6   | 0.1   |
|                                                                        | PrecMax     | 50.5   | 0.4   | 1.2   | 1.8   | 4.8   | 8.2   | 14.6  | 11.9  | 16.1  | 12.8  | 7.2   | 2.8   | 0.6   |
|                                                                        | PrecMin     | 26.6   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 2.2   | 1.6   | 2.1   | 2.4   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 6.7    | 0.1   | 0.3   | 0.4   | 1.1   | 1.9   | 3.3   | 2.6   | 3.3   | 2.7   | 1.7   | 0.8   | 0.2   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 92.9  | 96.7  | 99.3  | 101.4 | 100.8 | 95.4  | 93.5  | 93.2  | 91.7  | 91.8  | 91.0  | 91.4  |
|                                                                        |             | MCWB   | 70.4  | 71.8  | 71.3  | 72.9  | 73.6  | 75.4  | 75.8  | 75.6  | 76.1  | 75.9  | 73.5  | 70.6  |
|                                                                        | 2%          | DB     | 89.3  | 93.4  | 96.0  | 98.6  | 97.2  | 91.7  | 91.2  | 91.0  | 89.5  | 89.4  | 87.9  | 88.2  |
|                                                                        |             | MCWB   | 69.5  | 70.0  | 70.3  | 71.9  | 73.3  | 75.1  | 74.9  | 75.4  | 75.7  | 75.1  | 72.3  | 69.9  |
|                                                                        | 5%          | DB     | 85.1  | 89.9  | 93.0  | 95.4  | 94.8  | 89.2  | 88.7  | 88.3  | 87.5  | 86.8  | 85.4  | 85.8  |
|                                                                        |             | MCWB   | 68.3  | 69.7  | 70.2  | 71.5  | 73.3  | 74.6  | 74.7  | 75.1  | 75.1  | 74.3  | 71.5  | 69.9  |
|                                                                        | 10%         | DB     | 81.0  | 85.9  | 89.2  | 91.9  | 91.3  | 86.2  | 86.2  | 86.1  | 84.9  | 84.3  | 82.5  | 82.4  |
|                                                                        |             | MCWB   | 67.3  | 68.9  | 69.9  | 71.4  | 73.0  | 73.9  | 74.2  | 74.7  | 74.6  | 73.5  | 70.4  | 69.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 73.7  | 75.4  | 76.1  | 77.7  | 79.3  | 78.6  | 78.4  | 78.9  | 78.2  | 78.3  | 76.6  | 74.4  |
|                                                                        |             | MCDB   | 86.1  | 90.8  | 89.6  | 92.3  | 92.0  | 89.1  | 88.7  | 88.5  | 87.8  | 87.7  | 86.5  | 85.5  |
|                                                                        | 2%          | WB     | 71.4  | 72.8  | 73.6  | 75.2  | 76.5  | 76.6  | 76.7  | 77.2  | 76.8  | 76.5  | 74.1  | 72.1  |
|                                                                        |             | MCDB   | 84.6  | 87.6  | 88.0  | 90.8  | 90.0  | 87.7  | 87.7  | 87.5  | 86.5  | 86.5  | 83.7  | 83.3  |
|                                                                        | 5%          | WB     | 69.5  | 71.1  | 72.0  | 73.7  | 75.1  | 75.4  | 75.6  | 76.0  | 75.7  | 75.2  | 72.7  | 70.8  |
|                                                                        |             | MCDB   | 81.0  | 84.6  | 86.4  | 88.3  | 88.4  | 86.2  | 85.9  | 85.8  | 84.9  | 84.4  | 81.9  | 81.4  |
|                                                                        | 10%         | WB     | 68.1  | 69.7  | 70.8  | 72.6  | 74.0  | 74.5  | 74.6  | 74.9  | 74.7  | 74.0  | 71.5  | 69.6  |
|                                                                        |             | MCDB   | 78.1  | 81.5  | 84.6  | 86.5  | 86.3  | 84.0  | 84.0  | 84.0  | 83.1  | 82.1  | 80.0  | 79.4  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 24.5  | 26.6  | 27.2  | 26.6  | 23.7  | 19.4  | 19.7  | 19.7  | 18.1  | 18.6  | 21.0  | 23.3  |
|                                                                        |             | MCDBR  | 30.4  | 32.1  | 31.6  | 30.1  | 26.7  | 22.1  | 21.9  | 21.2  | 20.1  | 20.8  | 23.8  | 28.0  |
|                                                                        |             | MCWBR  | 13.5  | 13.1  | 11.6  | 10.5  | 9.2   | 8.6   | 9.2   | 9.1   | 8.4   | 9.2   | 10.7  | 12.4  |
|                                                                        | 5% WB       | MCDBR  | 27.9  | 29.5  | 28.7  | 27.2  | 23.9  | 20.0  | 20.6  | 20.3  | 18.9  | 19.3  | 21.6  | 25.6  |
|                                                                        |             | MCWBR  | 13.4  | 13.3  | 11.9  | 10.6  | 9.4   | 8.8   | 9.2   | 9.2   | 8.6   | 9.1   | 10.6  | 12.3  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.365 | 0.392 | 0.446 | 0.553 | 0.614 | 0.478 | 0.465 | 0.458 | 0.460 | 0.433 | 0.388 | 0.367 |
|                                                                        | taud        |        | 2.471 | 2.391 | 2.212 | 1.926 | 1.807 | 2.208 | 2.245 | 2.257 | 2.244 | 2.333 | 2.424 | 2.470 |
|                                                                        | Ebn at Noon |        | 293   | 290   | 276   | 247   | 229   | 260   | 264   | 268   | 268   | 273   | 283   | 289   |
|                                                                        | Edh at Noon |        | 34    | 38    | 47    | 62    | 69    | 46    | 44    | 44    | 45    | 40    | 35    | 33    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1358  | 1618  | 1851  | 1918  | 1773  | 1700  | 1865  | 1802  | 1584  | 1469  | 1405  | 1345  |
|                                                                        | RadStd      |        | 110   | 124   | 100   | 82    | 128   | 102   | 118   | 110   | 112   | 125   | 107   | 87    |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page